

Colour metallography of cast aluminium alloys

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Abstract

For better observation of the microstructure of cast aluminium alloys under an optical microscope, the contrast obtained for the different micro constituents was enhanced by making use of a copper phthalocyanine based colorant. The proposed method allows distinguishing structural details which would remain hidden or poorly resolved when studying the conventionally etched sample. The method produces high relief for silicon precipitates with the tint further accentuating the outlines, allowing detection of morphology change of the aluminium-silicon eutectic due to chemical modification with strontium. Magnesium micro segregations in the α -aluminium phase, caused by incorrect thermal treatment during ageing, can also be clearly distinguished. The coloured images obtained by optical microscopy are very sharp and provide similar or better information than the corresponding Scanning Electron Microscopy images. The method proved to be suitable for Quality Control in the fabrication of cast aluminium alloy compounds, sourcing down possible errors to different stages of the process.

Keywords

Coloured metallographic structures, optical microscopy, aluminium alloys, casting

Introduction

Before Electron Microscopes became commercially available in the late 1930's, visual metallographic investigations relied on optical microscope only, with its known limits in resolution power caused by the wavelength of the transmitted light. This often led to misinterpretations of phases or fine precipitates present in the microstructure. However, due to its low instrumental and running cost combined with the reduced need for sample preparation, optical microscopy is still one of the most important analytical techniques in metallography [1]. This is particularly true for cases where the composition and crystallography of phases possibly present in a given sample have already been predetermined making use of other techniques, such as X-ray diffraction, transmission electron microscopy or microprobe, in between others.

In order to improve the mechanical properties such as ductility, fracture, fatigue, impact properties and compositional homogeneity of as cast aluminium alloys, the strontium modification is possibly the most commonly applied chemical treatment [2,3,4]. It consists of modifying the morphology of faceted, brittle silicon plates in the unmodified eutectic to a fine fibrous microstructure. The mechanism is based on undercooling by chilling alone; the chemical modifier is required to accomplish this change at cooling rates achieved under normal foundry conditions for T6 temper [5]. In practice, strontium containing master alloy is added during a melt and tempering process. Under and over modification may occur during the process, resulting in the formation of undesirable compounds and enhanced porosity. The conditions for tempering and artificial ageing have also to be carefully chosen, as this process is to great extent responsible for the final material properties [6,7]. Quality Control for this process involves detection of porosity, eutectic silicon needle or fibrous structure, undesired compounds such as SrSi_2 or $\text{Al}_2\text{Si}_2\text{Sr}$ or segregated phases, e.g. Al_2Cu . In particular the first target features should be suitable for detection by optical microscopy when the sample is adequately prepared.

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Meanwhile features like cracks, porosity or inclusions may be visible under optical microscopy on a simply polished sample; the investigation on microstructural details such as grain boundaries, crystal orientation, size, segregation, phase distribution and precipitates normally requires etching [8]. This is a process of selective corrosion of microstructural features, causing differences in topography or in the reflectivity of the different phases. Kinetics of this attack are driven by the crystallographic orientation and the process will also preferentially etch high energy sites such as grain boundaries, resulting in a surface relief allowing distinguishing the before mentioned features. Standard chemical etchants are a mixture of a corrosive agent (acid), a modifier that reduces ionization (e.g. alcohol) and an oxidizing component (e.g. hydrogen peroxide). Optimum concentration and application time strongly depend on the targeted microstructure and the overall composition of the sample matrix.

In order to reveal features in metallic microstructures that cannot be resolved when applying standard sample preparation procedures, the tailor made use of special purpose etchants [9] has always been an ongoing field of investigation in metallography. These may be categorised in three main groups: (a) delineating etchants revealing the general structure and usually based on an acidic attack, (b) electrolytic etchants revealing either the general structure or are specific to a certain phase and (c) staining or film forming etchants, generally of alkaline nature and raising colour in specific phases.

The metallographic examination of stained microstructures in metals and alloys can significantly improve their observation by optical microscopy. Many important details for correct identification of phases, impurities, defects and segregations present in a microstructure, which on grey scale would be undetectable, can be revealed in this way.

Most of research carried out up to date on coloured metallography is making use of staining etchants, generally sodium or potassium hydroxide with an oxidising agent such as picrates, permanganates or ferricyanides added, which produce coloured reaction products on the metallic surface [10,11].

For wrought and cast aluminium alloys, staining etchants in optical metallography are widely used [12,13]. Aim is to achieve a selective staining in distinguishable colours for the different intermetallic compounds and segregations upon their mayor content in elements such as copper, iron, manganese etc., the latter able to form intensively coloured salts [14]. The binary aluminium-silicon alloy for casting, grade ENAC AlSi7Mg0.6, is a material subjected to a thermal treatment following Standard EN 1706 : 2010, which incorporates 0.3 -0.6 % in weight of magnesium into the alloy. In this particular case, classical staining etchants are poorly efficient.

The approach followed in the present investigation to overcome this drawback is starting up with a deep chemical attack of the metals surface, exposing it to a concentrated aqueous solution of sodium hydroxide in an ultrasonic bath. Afterwards a phthalocyanine blue based lacquer is applied, which has been previously found to be suitable for investigating quenching structures and isothermal phase transformation of carbon iron alloys [15]. The proposed method is evaluated for detection of microstructural features resulting from faulty chemical or heat treatment of as cast aluminium alloys during the industrial production process.

Experimental

Sample and reagents

The alloy selected in this investigation is EN AC-42200 (EN AC-AlSi7Mg0,6), subjected to normative EN 1706 : 2010. Ingots and processed items were provided by the manufacturer of aluminium alloy elements Ditalen S.L. (Spain). The chemical composition of the material is summarised in Table1:

Table 1: Chemical composition of the EN AC-42200 alloy (in weight %, remainder aluminium)

Si	Fe	Cu	Mn	Mg	Cr	Ni	Zn	Pb	Sn	Ti
6,5-7,5	0,19	0,05	0,1	0,45-0,7	-	-	0,07	-	-	0,08-0,25

NaOH pellets and 37% HCl were both analytical grade and obtained from Carlo Erba, (Milan, Italy), Deionized water to prepare diluted alkaline solutions was obtained from a Millipore (Bedford, MA,

USA) ZMFQ 23004 Milli-Q[®] water system. “Cinco Aros” Zapon lacquer was purchased from Mongay S.A. (Barcelona, Spain).

Figure 1 shows a range of the machined parts under study as delivered, some of them already sampled. They have been mould casted and tempered by Ditalen S.L., are designated to railroad high-speed operation components and hence need matching highest standards in fabrication and Quality Control in order to avoid high profile failures.



Figure 1: Examples for raw material and workpieces made from cast aluminium alloy, (a) AISi7Mg0,6 ingot (b) anchor clamp, (c), part of registration device, (d) tensioning wheel for overhead catenary system

Tint preparation

Details on the dye chosen are given elsewhere [15], briefly it is a 3% (M/V) solution of phtalocyanine blue in Zapon lacquer. The latter is a nitrocellulose based varnish in a solvent mixture containing ethanol, butane-2-ol, xylene, toluene, n-butyl- and pentylacetate. The reason for the use of this particular varnish as carrier for the dye is that it provides high dispersant and binding power, is transparent, able to form very thin films and dries off fast.

Sample preparation

After cutting samples of appropriate size out of the original specimen, a conventional grinding and polishing process, consisting in coarse grinding on a belt grinder (Buehler Metaserv 3000), followed by wet grinding on increasingly finer grade round grinding papers (Buehler-Met 320, 600 and 2000 abrasive discs). The samples were then polished with Aluminium Oxide powder having a particle diameter of 0.3 μm (Buehler-Met Alumina Suspension, Alpha Micropolish Alumina-1). Subsequently, etching and staining sample preparation was carried out in two steps:

- Deep etching of the materials surface with an aqueous dissolution of 30% NaOH during 5 min in an ultrasonic bath. The sample was then thoroughly washed with distilled water and exposed to a short, 1 min cleaning step making use of slightly acidified water (0.015% hydrochloric acid) in the ultrasonic bath. Thereafter the sample was washed subsequently with distilled water and ethanol and dried.
- Immediately prior observation under the optical microscope, the tint was brought onto the samples surface using a smooth painters brush. In just a few seconds at ambient temperature the solvent dries off, a stable film forms and the sample is ready for investigation.

For samples investigated under SEM, step b was skipped. As the material is conductive, an additional sputtering process was not necessary. Sample size was chosen to be suitable for direct mounting onto the sample holder of the SEM. Figure 2 shows a sample which had gone through the two step etching and staining sample preparation process.



Figure 2: A sample readily prepared for observation under optical microscopy

Instrumental

Optical metallographic investigations were carried out on a Zeiss Axiovert A1 Inverted Phase Contrast Microscope (Carl Zeiss, Jena, Germany). A Scanning Electron Microscope Jeol JM-6400 (Tokyo, Japan) was employed for unstained samples.

Results and Discussion

The observed microstructures, resulting from casting, brazing, welding, heat treating or mechanical working of the alloy, may be in our case categorised into the different steps of fabrication and possible errors herein incurred (Temper designations in parenthesis):

- structure of the original ingot
- structure of the mould casting, solution heat treated with strontium and artificially aged (T6)
- structure of the casting, faulty heat treatment with strontium and artificially aged (T6)
- structure of the casting, solution heat treated with strontium and faulty artificial ageing (T6)

The microstructure of the original ingot shows typical dendrite growth morphologies for Al-Si alloys, consisting of primary α -phase aluminium dendrites and a eutectic interdendritic phase with irregular silicon crystals of needle like, feathered shape. Within the primary α -phase dendrites, Mg_2Si intermetallic particles in Widmanstätten structure can be observed. These details are shown in Figures 3 and 4 for observation under the Scanning Electron Microscope for an etched sample and with Optical Microscopy for an etched and stained sample, respectively.

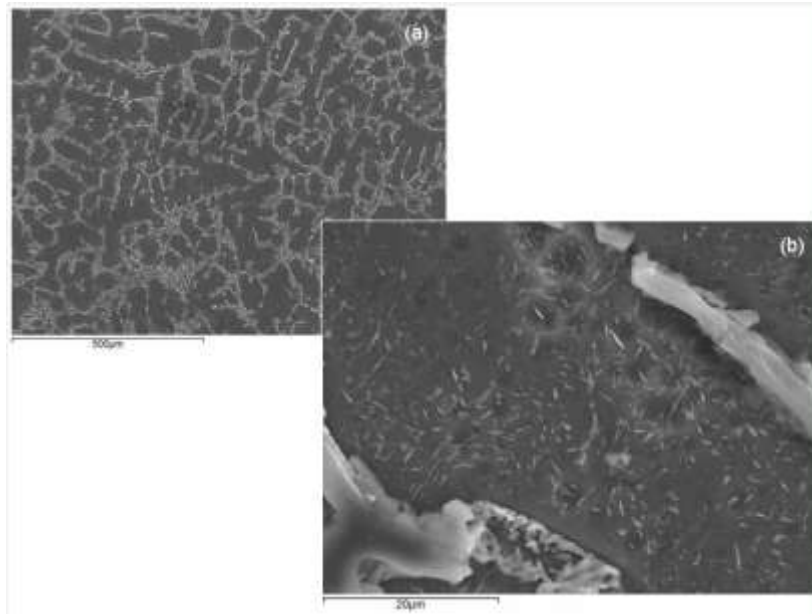


Figure3: SEM image of an etched sample taken from the original ingot at low (a) and higher (b) magnification. A dendritic structure of α phase (aluminum) and eutectic silicon crystals in interdendritic spaces can be distinguished. Mg_2Si crystals are precipitated in the dendrites of the α phase.

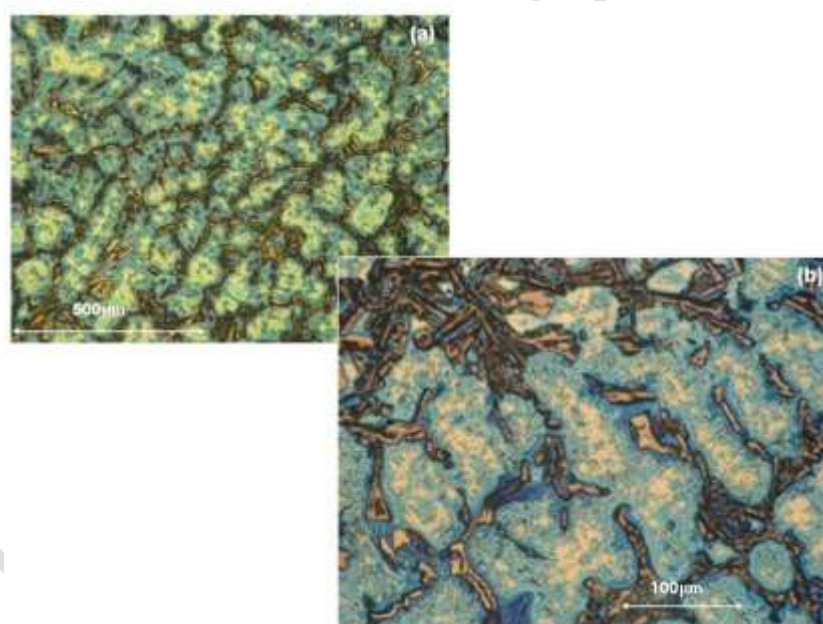


Figure 4: Optical micrographs of an etched ingot sample stained with phtalocyanine blue, following the proposed methodology at low (a) and higher (b) magnification.

After the sequence of casting operations for strontium modification in EN AC-42200 type alloys, temper designation T6 conditions have to be reached. A correct T6 thermal treatment involves in this case a solubilisation step carried out at a temperature around 500 °C and subsequently quenching to near room temperature, followed by artificial ageing at about 150 °C for several hours. As shown in figures 5 and 6, the eutectic silicon morphology has clearly changed from acicular, feathered shape to the desired, more rounded and fibrous microstructure. The Mg_2Si crystals in the dendrites of the α phase are also now absent, all that indicating that the heat treatment was correct.

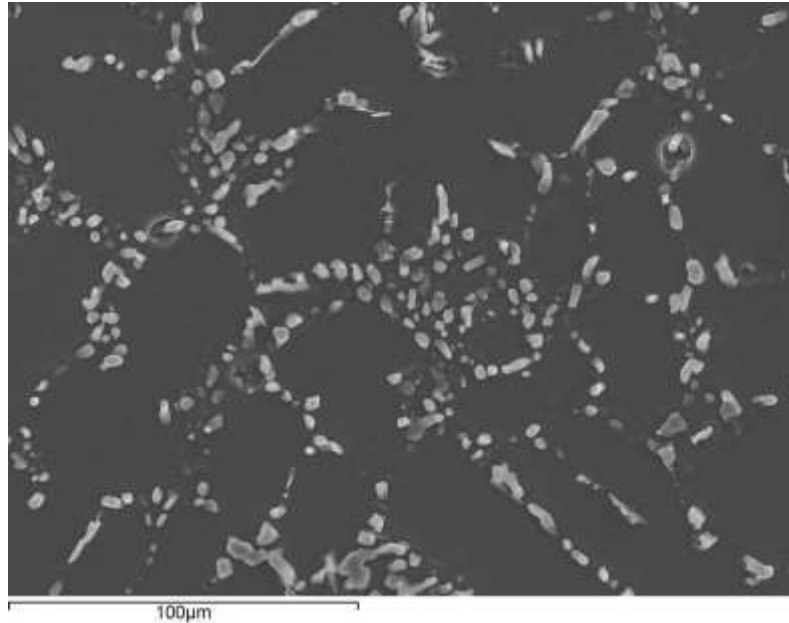


Figure.5: SEM image of the alloy under study after correct strontium modification and artificial ageing (Temper designation T6). Roundish shaped and smaller eutectic silicon and the absence of Mg_2Si crystals in the α phase dendrites can be observed.

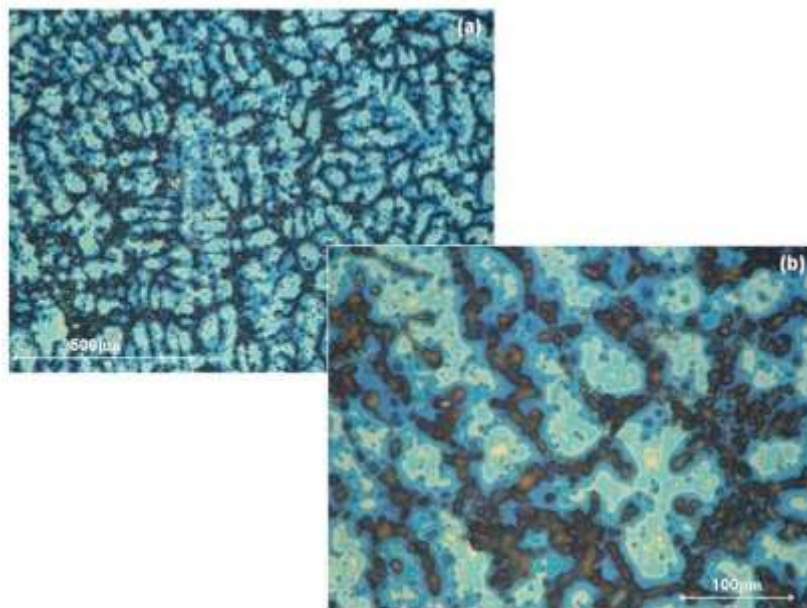


Figure.6: Optical micrographs at two different magnifications of the structure described in Figure 5 after staining the sample with phtalocyanine blue. (b) is comparable to the magnification applied for obtaining Fig. 5 in SEM.

Ideally, a strontium content of about 0.01-0.015 weight % is sought in the final alloy [16]. In practise, this goal is not always achieved and under and over modification can occur; being the latter preferable if the control of the strontium level is not good enough. In particular too low strontium concentrations will lead to incomplete transformation of the acicular eutectic silicon, even though the subsequent heat treatment to reach T6 conditions is carried out correctly. This is the case for the microstructures shown in figures 7 and 8. It can be observed that the interdendritic silicon is not well globulised, whereas the intermetallic Mg_2Si crystals are absent, indicating effective heat treatment. This may lead to a final product of correct hardness, but with values for tensile strength, elongation or density out of specifications.

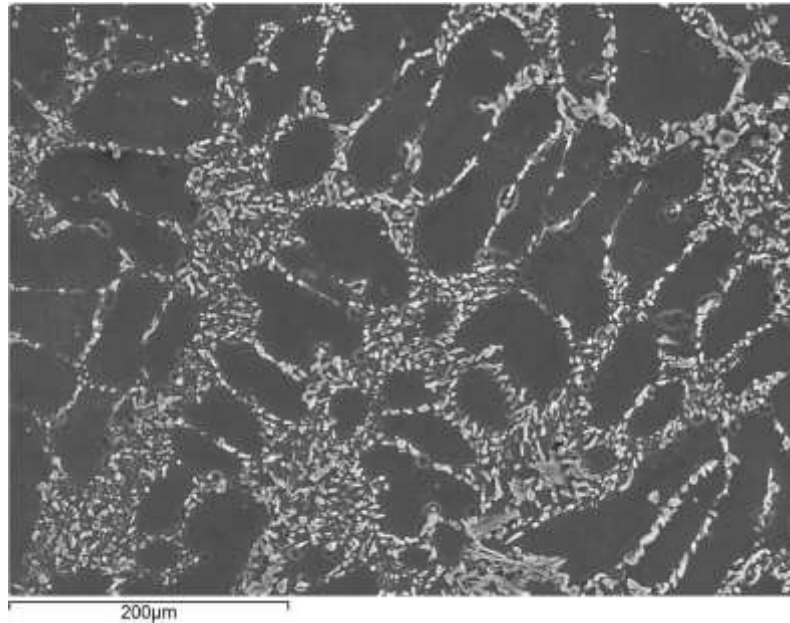


Figure 7: SEM image of a sample with failed strontium modification and correct heat treatment. The morphology and size of eutectic silicon crystals in the interdendritic space is irregular. There are no Mg_2Si crystals precipitated in the dendrites of the α Al-phase.

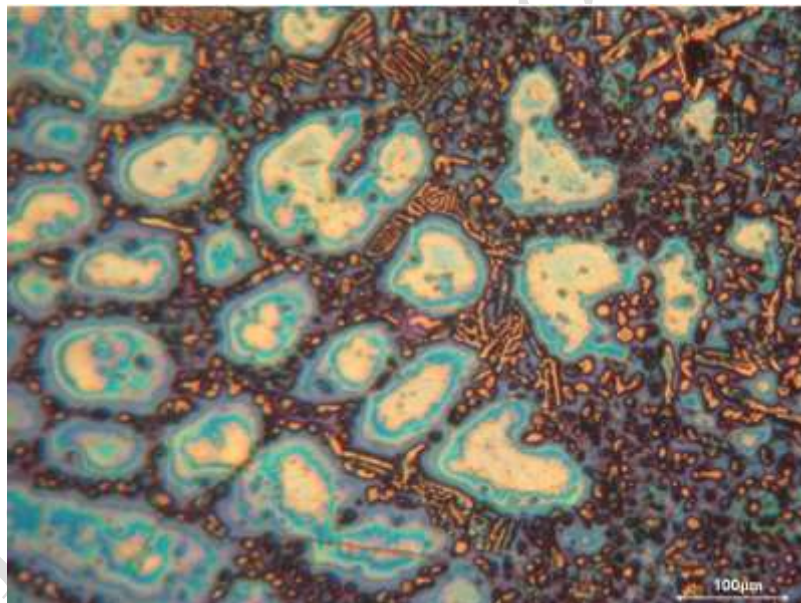


Figure 8: Optical micrograph of a stained sample with failed strontium modification and correct heat treatment.

During the mould casting process of the alloy under study, there is also the possibility that the strontium modification did work properly but the heat treatment applied to reach T6 conditions is nonconforming. This becomes manifest by the presence of finely recrystallised Mg_2Si within the dendrites of the α aluminium-phase, acting as stress factor and impairing the mechanical properties of the alloy. In spite of the fact that the highly alkaline etching process applied will dissolve these crystals, the surface marking left in the α -aluminium matrix reveals their presence, as shown in Figure 9. When observing the stained samples under an optical microscope (Figure 10), this effect appears as a sort of segregation in the interior of the dendrites of α aluminium phase, which the preferred site for intermetallic precipitation.

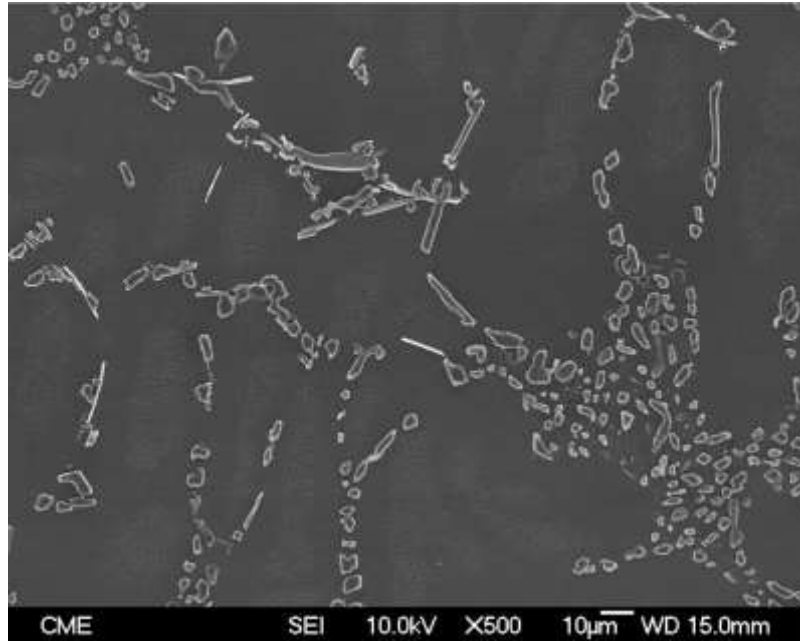


Figure.9: SEM image of a sample with successful strontium modification and failed heat treatment. The voids left by Mg_2Si crystals redissolved during chemical etching can be clearly distinguished.

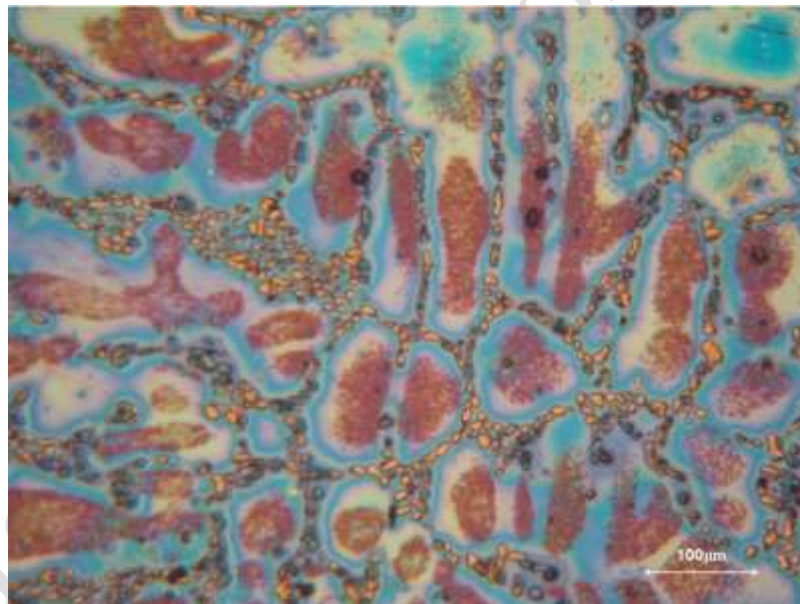


Figure.10: Optical micrograph of a sample with successful strontium modification and failed heat treatment after staining with phtalocyanine blue.

The metallographic sample preparation proposed in this work is also very efficient in the detection of micro porosity caused by volume contraction during the solidification process. When the thin lacquer film is applied after etching of the sample, capillary action causes the tint to be absorbed into these micro pores, leaving the borders in unstained condition (Figure 11). This significantly enhances the contrast for the outline of these pores and provides, at low magnifications, an excellent and intuitive visual too for the estimation of extend and shape of pores present in the sample.

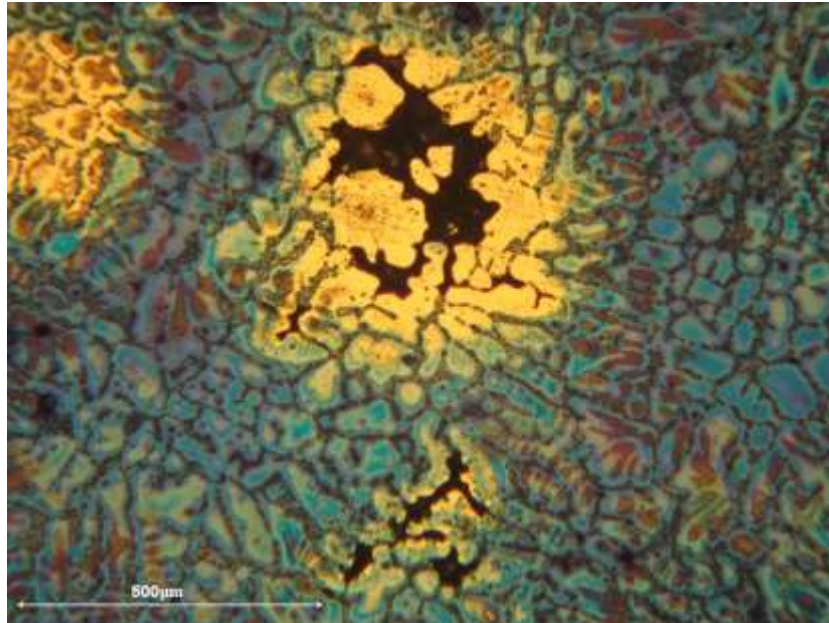


Figure.11: Optical micrograph, in which a dye free zone around a highly irregular micropore can be observed.

Conclusions

A simple and fast, two step metallographic sample preparation method is proposed for cast aluminium alloys. Etching is carried out in highly alkaline conditions, followed by staining with a phtalocyanine blue based dye.

Important parameters affecting Quality Control of the finished aluminium alloy work pieces, such as crystal size and orientation, segregation, phase distribution and porosity can be observed by simple optical microscopy, providing a cost effective and high throughput method which is accessible even to low profile laboratories. For the chosen set of parameters, Scanning Electron Microscopy did not yield additional or more precise information. The observed characteristics even allow associating undesired microstructures to particular steps of the fabrication process, such as unsuccessful strontium modification or faulty tempering.

The metallographic preparation proposed for AlSi7Mg0,6 (EN AC-42200) should also be, without further modifications, applicable to other cast aluminium alloys of similar structure, such as AlSi10Mg (EN AC-43000), AlSi5Cu (EN AC-45000), AlSi9Cu (EN AC-46000), AlSi(Cu) (EN AC-47000) or AlSiCuNiMg (EN AC-48000).

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